



WEB-BASED APPLICATION
DEVELOPMENT
(CSE 3023)

Lab 7

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Date: 12/5/2026

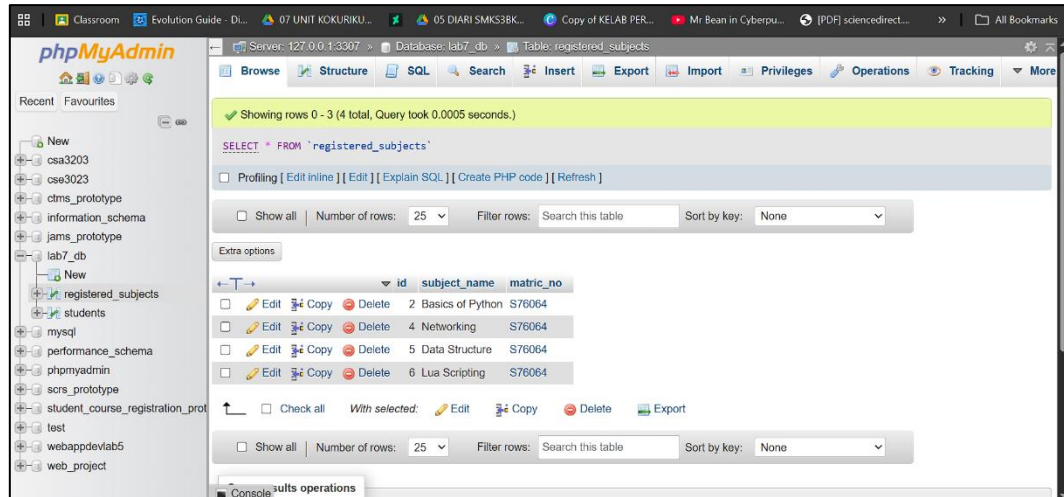
Group: K1

Program: Computer Science (Software Engineering)

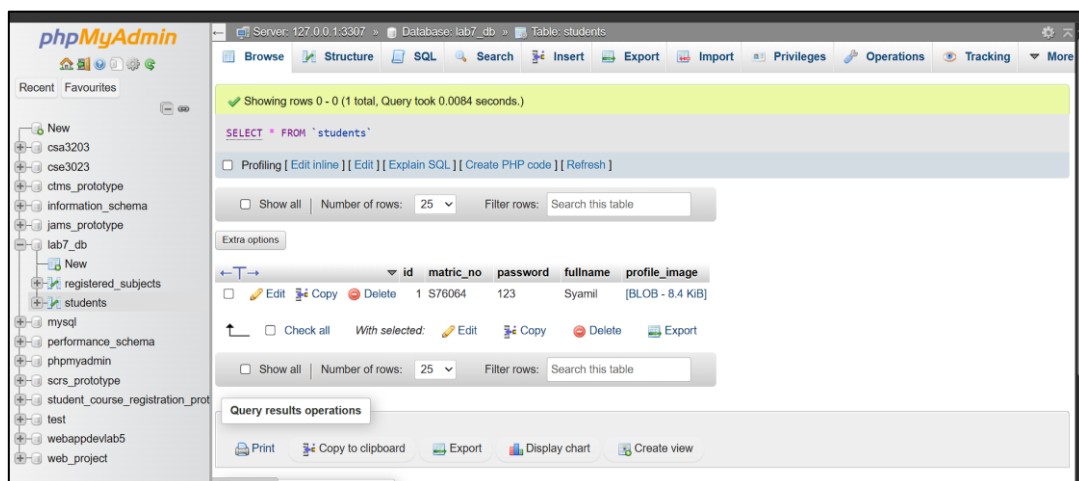
Lab 7 Exercise

PHP MyAdmin:

registered subjects table



students table



Source Code (JSP):

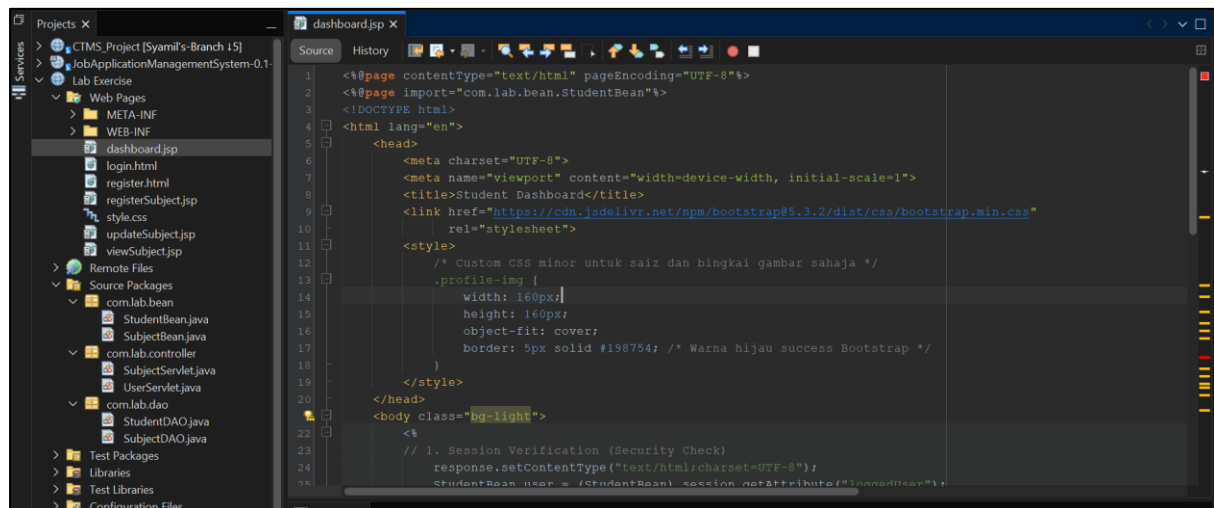
dashboard.jsp

```

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<%@page import="com.lab.bean.StudentBean"%>
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1">
    <title>Student Dashboard</title>
    <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstra
p.min.css"
    rel="stylesheet">
    <style>
      /* Custom CSS minor untuk saiz dan bingkai gambar sahaja
*/
      .profile-img {
        width: 160px;
        height: 160px;
        object-fit: cover;
        border: 5px solid #198754; /* Warna hijau success
Bootstrap */
      }
    </style>
  </head>
  <body class="bg-light">
    <%
      // 1. Session Verification (Security Check)
      response.setContentType("text/html; charset=UTF-8");
      StudentBean user = (StudentBean)
session.getAttribute("loggedUser");
      if (user == null) {
        response.sendRedirect("login.html");
        return;
      }
    %>
    <div class="container d-flex justify-content-center align-
items-center"
      style="min-height: 100vh;">
      <div class="card shadow-lg border-0 rounded-4 p-4 p-md-5
text-center"
        style="max-width: 600px; width: 100%;">
        <div class="card-body">
          <h2 class="fw-bold text-dark mb-
2">Welcome, <%= user.getFullname() %>!/></h2>
        <p class="text-muted fs-5 mb-4">Matric Number:
<%=user.getMatricNo()%> <strong class="text-secondary"><div
class="d-flex justify-content-center gap-3 flex-wrap">
        <a href="UserServlet?action=logout"
class="btn btn-dark px-4 py-2
        fw-semibold">Logout</a>
        <a href="UserServlet?action=delete"
        onclick="return confirm('WARNING:
Are you sure you want to delete
        your account permanently?');"
        class="btn btn-danger px-4 py-2
fw-semibold">Delete Account</a>
        </div>
    </div>
</div>
</div>
<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.
bundle.min.js"></script>
</body>
</html>

```

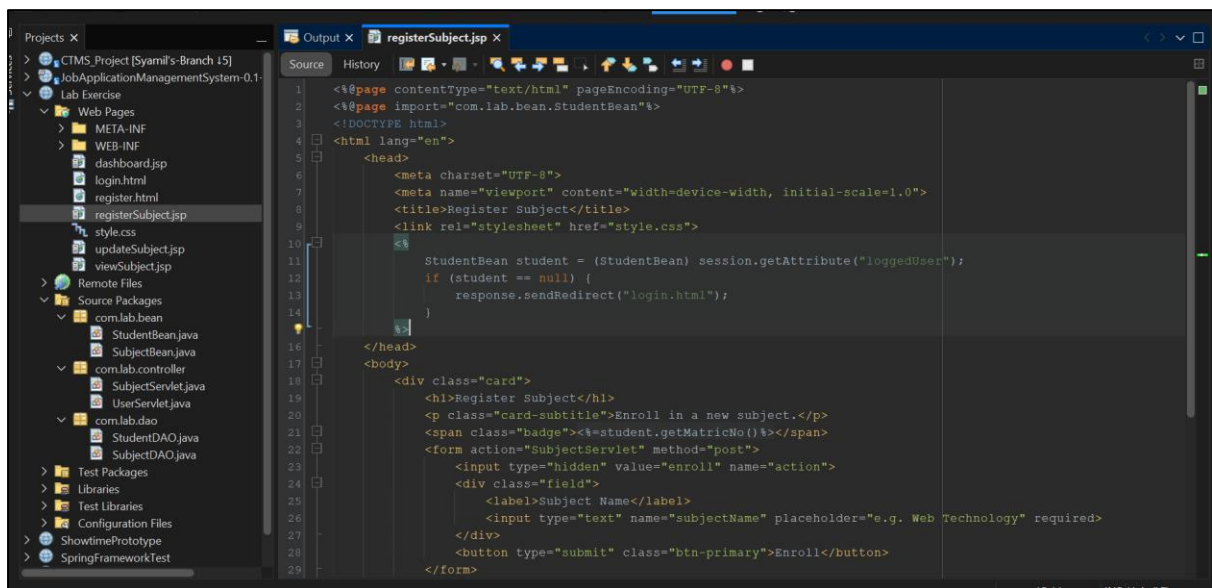
Netbean



registerSubject.jsp

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<%@page import="com.lab.bean.StudentBean"%>
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Register Subject</title>
    <link rel="stylesheet" href="style.css">
    <%
        StudentBean student = (StudentBean)
session.getAttribute("loggedUser");
        if (student == null) {
            response.sendRedirect("login.html");
        }
    %>
</head>
<body>
    <div class="card">
        <h1>Register Subject</h1>
        <p class="card-subtitle">Enroll in a new subject.</p>
        <span class="badge"><%=student.getMatricNo()%></span>
        <form action="SubjectServlet" method="post">
            <input type="hidden" value="enroll" name="action">
            <div class="field">
                <label>Subject Name</label>
                <input type="text" name="subjectName" placeholder="e.g. Web
Technology" required>
            </div>
            <button type="submit" class="btn-primary">Enroll</button>
        </form>
    </div>
</body>
</html>
```

Netbean



updateSubject.jsp

```

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<%@page import="java.util.*, com.lab.bean.*" %>
<!DOCTYPE html>
<html lang="en">
    <head>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-
scale=1.0">
        <title>Edit Subject</title>
        <link rel="stylesheet" href="style.css">
        <%
            SubjectBean subject = (SubjectBean)
request.getAttribute("subject");
        %>
    </head>
    <body>
        <div class="card">
            <h1>Edit Subject</h1>
            <p class="card-subtitle">Update or remove this
subject.</p>
            <form action="SubjectServlet" method="post">
                <input type="hidden" name="action" value="update">
                <input type="hidden" name="id"
value="<%=subject.getId() %>">
                <div class="field">
                    <label>Subject Name</label>
                    <input type="text" name="subjectName"
value="<%=subject.getSubjectName() %>">

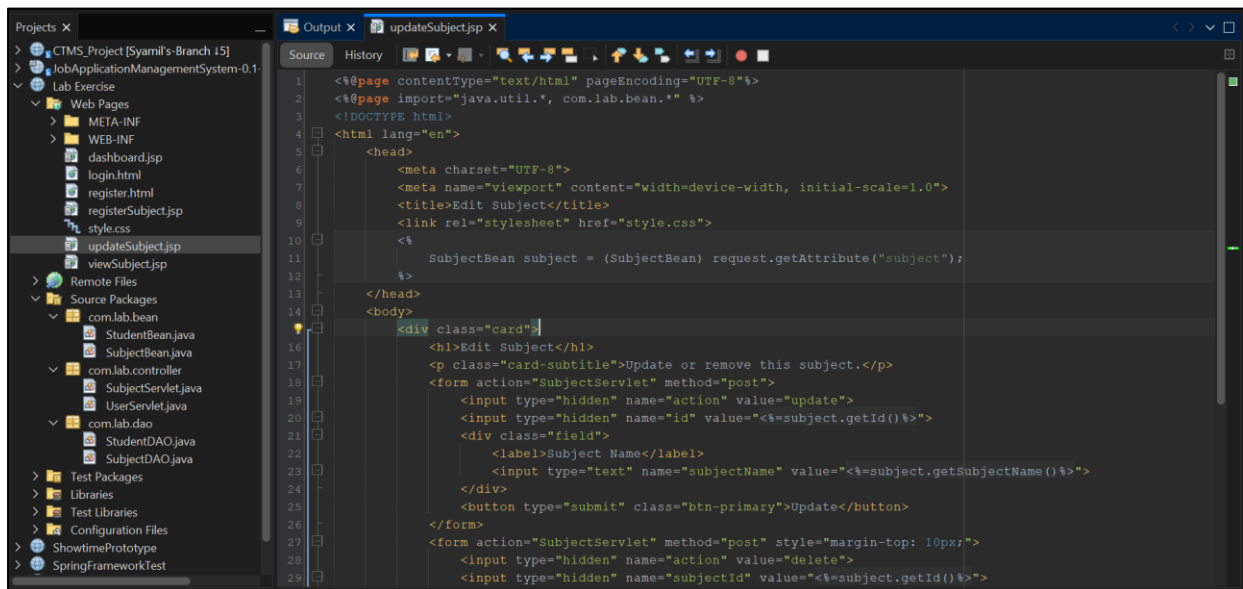
```

```

        </div>
        <button type="submit" class="btn-
primary">Update</button>
    </form>
    <form action="SubjectServlet" method="post"
style="margin-top: 10px;">
        <input type="hidden" name="action" value="delete">
        <input type="hidden" name="subjectId"
value="<%=subject.getId() %>">
        <button type="submit" class="btn-danger"
style="width:100%">Delete Subject</button>
    </form>
</div>
</body>
</html>

```

Netbean



viewSubject.jsp

```

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<%@page import="java.util.*, com.lab.bean.*" %>
<!DOCTYPE html>
<html lang="en">
    <head>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-
scale=1.0">
        <title>Subject List</title>
        <link rel="stylesheet" href="style.css">
    <%

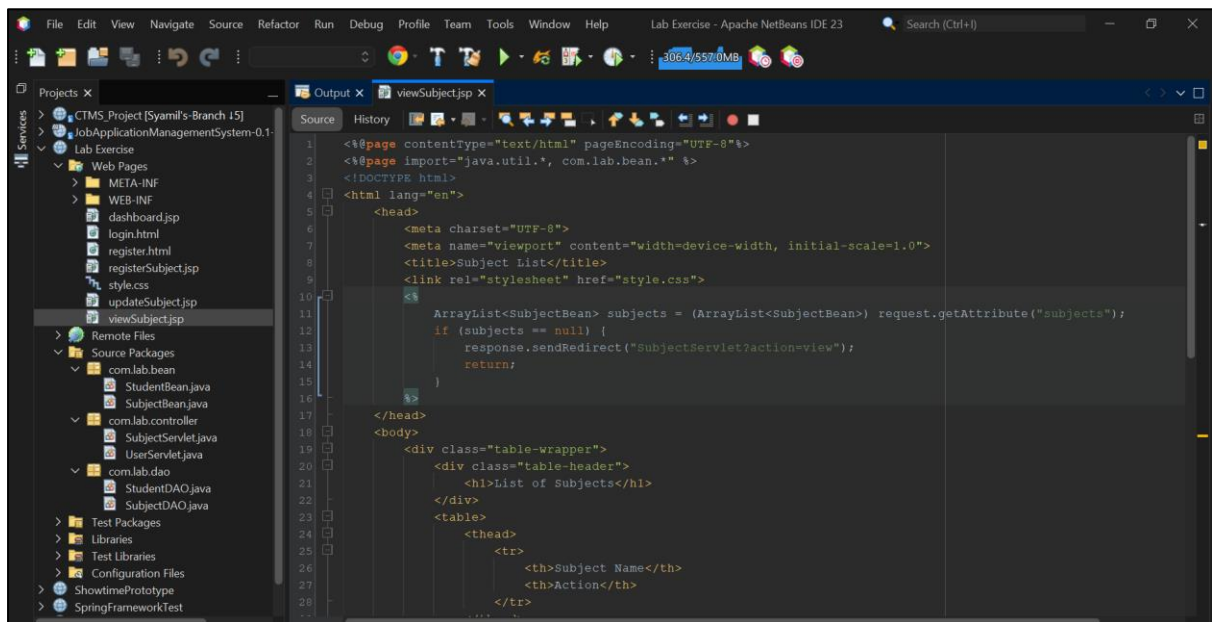
```

```

        ArrayList<SubjectBean> subjects =
(ArrayList<SubjectBean>) request.getAttribute("subjects");
        if (subjects == null) {
            response.sendRedirect("SubjectServlet?action=view");
            return;
        }
    %>
</head>
<body>
    <div class="table-wrapper">
        <div class="table-header">
            <h1>List of Subjects</h1>
        </div>
        <table>
            <thead>
                <tr>
                    <th>Subject Name</th>
                    <th>Action</th>
                </tr>
            </thead>
            <tbody>
                <%
                    for (SubjectBean sb : subjects) {
                %>
                <tr>
                    <td><%=sb.getSubjectName() %></td>
                    <td>
                        <button class="btn-secondary"
                            onclick="location.href =
'SubjectServlet?action=viewUpdate&subjectId=<%=sb.getId() %>'>
                            Edit
                        </button>
                    </td>
                </tr>
                <%
                    }
                %>
            </tbody>
        </table>
    </div>
</body>
</html>

```


Netbean



Source Code (HTML):

login.html

```

<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Student Login</title>
    <link rel="stylesheet" href="style.css">
  </head>
  <body>
    <div class="card">
      <h2>Student Login</h2>
      <p class="card-subtitle">Enter your credentials to
continue.</p>
      <form action="UserServlet" method="POST">
        <input type="hidden" name="action" value="login">
        <div class="field">
          <label for="matricNo">Matric No</label>
          <input type="text" id="matricNo" name="matricNo"
placeholder="e.g. 2023001234" required>
        </div>
        <div class="field">
          <label for="password">Password</label>
          <input type="password" id="password"

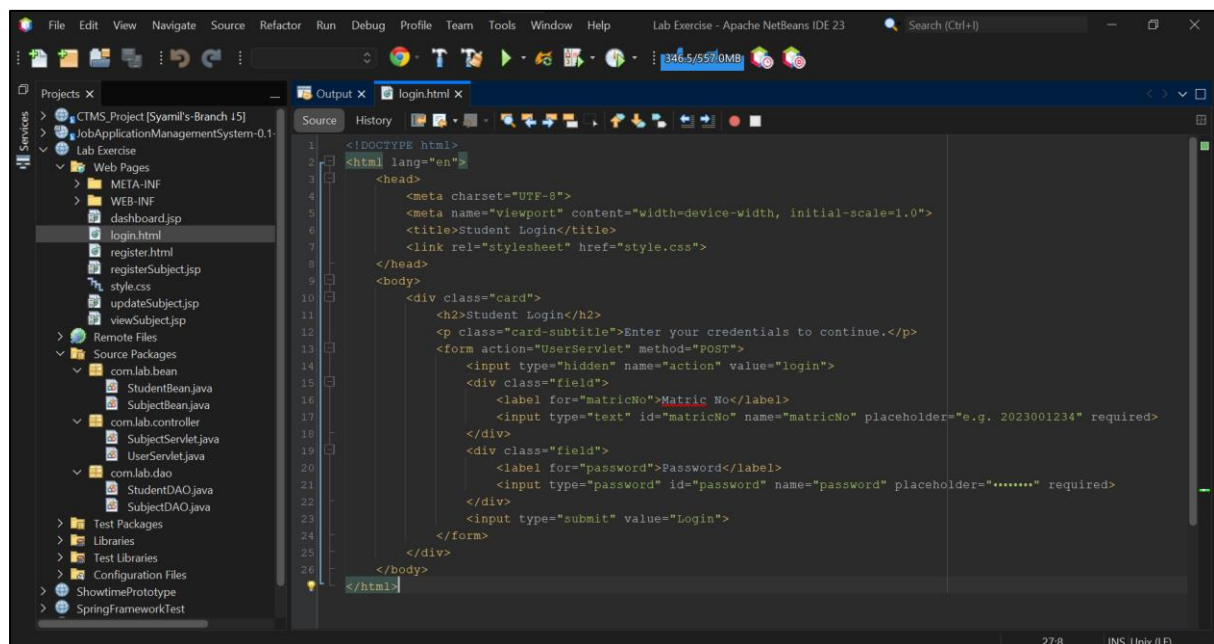
```

```

name="password" placeholder="....." required>
    </div>
    <input type="submit" value="Login">
  </form>
</div>
</body>
</html>

```

Netbean



register.html

```

<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Student Registration</title>
    <link rel="stylesheet" href="style.css">
  </head>
  <body>
    <div class="card">
      <h2>Register New Student</h2>
      <p class="card-subtitle">Create your student
account.</p>
      <form action="UserServlet" method="POST"

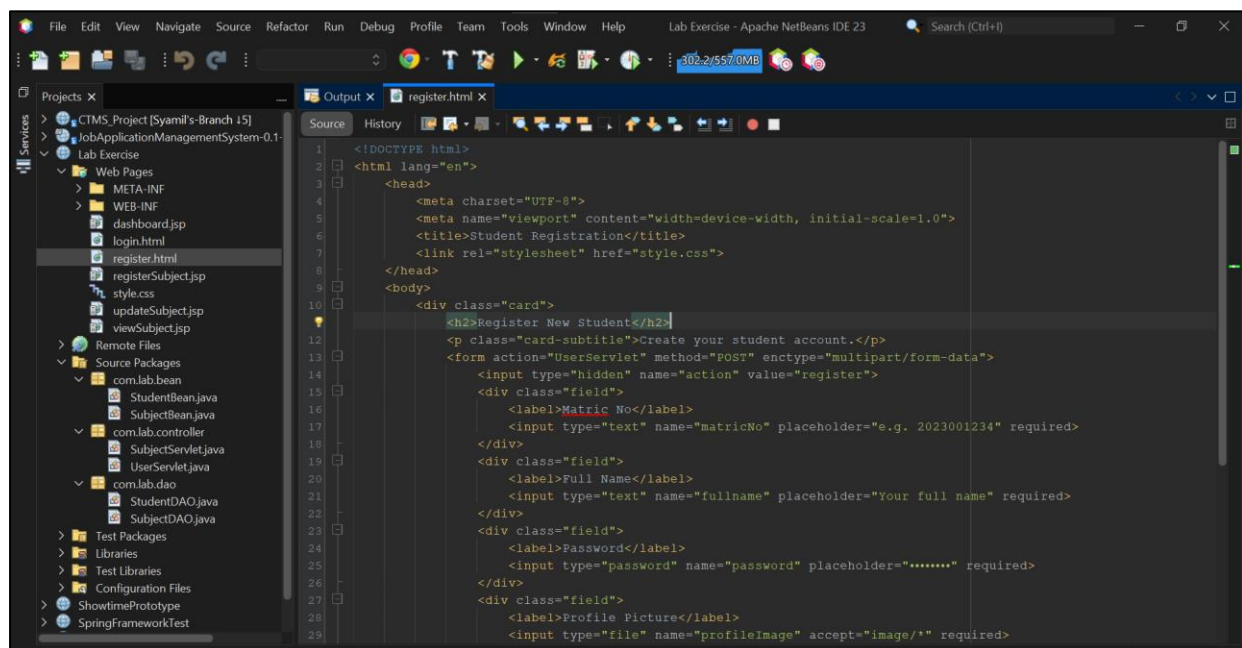
```

```

enctype="multipart/form-data">
    <input type="hidden" name="action" value="register">
    <div class="field">
        <label>Matric No</label>
        <input type="text" name="matricNo"
placeholder="e.g. 2023001234" required>
    </div>
    <div class="field">
        <label>Full Name</label>
        <input type="text" name="fullname"
placeholder="Your full name" required>
    </div>
    <div class="field">
        <label>Password</label>
        <input type="password" name="password"
placeholder="....." required>
    </div>
    <div class="field">
        <label>Profile Picture</label>
        <input type="file" name="profileImage"
accept="image/*" required>
    </div>
    <input type="submit" value="Register">
</form>
</div>
</body>
</html>

```

Netbean



Source Code (CSS):

style.css

```

@import
url('https://fonts.googleapis.com/css2?family=IBM+Plex+Sans:wght@400
;500;600&family=IBM+Plex+Mono:wght@400;500&display=swap');

:root {
  --bg:          #f4f5f7;
  --surface:     #ffffff;
  --border:      #dde1e7;
  --text:        #1a1d23;
  --muted:       #6b7280;
  --accent:      #2563eb;
  --accent-h:    #1d4ed8;
  --danger:      #dc2626;
  --danger-h:    #b91c1c;
  --success:     #16a34a;
  --radius:      8px;
  --shadow:       0 1px 3px rgba(0,0,0,.08), 0 1px 2px rgba(0,0,0,.05);
  --shadow-md:    0 4px 12px rgba(0,0,0,.10);
}

/* — Reset & Base ————— */
*, *::before, *::after { box-sizing: border-box; margin: 0; padding: 0; }

body {
  font-family: 'IBM Plex Sans', sans-serif;
  background: var(--bg);
  color: var(--text);
  font-size: 15px;
  line-height: 1.6;
  min-height: 100vh;
  display: flex;
  align-items: center;
  justify-content: center;
  padding: 32px 16px;
}

/* — Card Container ————— */
.card {
  background: var(--surface);
  border: 1px solid var(--border);
  border-radius: var(--radius);
  box-shadow: var(--shadow-md);
  width: 100%;
  max-width: 440px;
  padding: 36px 40px;
}

```

```

}

/* — Page Title / Header — */
h1, h2 {
  font-size: 1.4rem;
  font-weight: 600;
  color: var(--text);
  margin-bottom: 4px;
}

.card-subtitle {
  font-size: 0.875rem;
  color: var(--muted);
  margin-bottom: 28px;
}

/* Divider under heading */
h1 + form, h2 + form { margin-top: 24px; }

/* — Form Layout — */
form { display: flex; flex-direction: column; gap: 16px; }

.field {
  display: flex;
  flex-direction: column;
  gap: 5px;
}

label {
  font-size: 0.8125rem;
  font-weight: 500;
  color: var(--muted);
  letter-spacing: .02em;
  text-transform: uppercase;
}

input[type="text"],
input[type="password"],
input[type="file"] {
  width: 100%;
  padding: 9px 12px;
  border: 1px solid var(--border);
  border-radius: var(--radius);
  font-family: inherit;
  font-size: 0.9375rem;
  color: var(--text);
  background: var(--surface);
  transition: border-color .15s, box-shadow .15s;
  outline: none;
}

```

```

input[type="text"]:focus,
input[type="password"]:focus,
input[type="file"]:focus {
  border-color: var(--accent);
  box-shadow: 0 0 0 3px rgba(37,99,235,.12);
}

input[type="file"] {
  cursor: pointer;
  padding: 7px 12px;
  font-size: 0.875rem;
  color: var(--muted);
}

/* — Buttons — */
button, input[type="submit"] {
  display: inline-flex;
  align-items: center;
  justify-content: center;
  gap: 6px;
  padding: 9px 18px;
  border: none;
  border-radius: var(--radius);
  font-family: inherit;
  font-size: 0.9375rem;
  font-weight: 500;
  cursor: pointer;
  transition: background .15s, transform .1s;
}

input[type="submit"],
button[type="submit"],
.btn-primary {
  background: var(--accent);
  color: #fff;
  width: 100%;
  margin-top: 4px;
}

input[type="submit"]:hover,
button[type="submit"]:hover,
.btn-primary:hover {
  background: var(--accent-h);
}

input[type="submit"]:active,
button[type="submit"]:active {
  transform: scale(.98);
}

.btn-secondary {

```

```

    background: var(--bg);
    color: var(--text);
    border: 1px solid var(--border);
}

.btn-secondary:hover { background: var(--border); }

.btn-danger {
    background: var(--danger);
    color: #fff;
}

.btn-danger:hover { background: var(--danger-h); }

/* — Table (Subject List) ————— */
.table-wrapper {
    width: 100%;
    max-width: 680px;
    background: var(--surface);
    border: 1px solid var(--border);
    border-radius: var(--radius);
    box-shadow: var(--shadow-md);
    overflow: hidden;
}

.table-header {
    padding: 20px 24px 12px;
    border-bottom: 1px solid var(--border);
}

table {
    width: 100%;
    border-collapse: collapse;
    font-size: 0.9375rem;
}

thead tr {
    background: var(--bg);
}

th {
    text-align: left;
    padding: 10px 24px;
    font-size: 0.8rem;
    font-weight: 600;
    letter-spacing: .04em;
    text-transform: uppercase;
    color: var(--muted);
    border-bottom: 1px solid var(--border);
}

```

```

td {
  padding: 12px 24px;
  border-bottom: 1px solid var(--border);
  color: var(--text);
}

tbody tr:last-child td { border-bottom: none; }

tbody tr:hover { background: #f9fafb; }

/* Action buttons inside table */
td button {
  padding: 5px 14px;
  font-size: 0.8125rem;
  width: auto;
  margin-top: 0;
}

/* — Info Badge (matric no) ————— */
.badge {
  display: inline-block;
  background: var(--bg);
  border: 1px solid var(--border);
  border-radius: 20px;
  padding: 3px 12px;
  font-family: 'IBM Plex Mono', monospace;
  font-size: 0.8125rem;
  color: var(--muted);
  margin-bottom: 20px;
}

/* — Alert / Flash messages ————— */
.alert {
  padding: 10px 14px;
  border-radius: var(--radius);
  font-size: 0.875rem;
  border-left: 3px solid;
}

.alert-error { background: #fef2f2; border-color: var(--danger);
color: #991b1b; }
.alert-success { background: #f0fdf4; border-color: var(--success);
color: #166534; }
.alert-info { background: #eff6ff; border-color: var(--accent);
color: #1e40af; }

/* — Utility ————— */
.text-muted { color: var(--muted); font-size: 0.875rem; }
.mt-8 { margin-top: 8px; }
.mt-16 { margin-top: 16px; }
.gap-8 { display: flex; gap: 8px; }

```

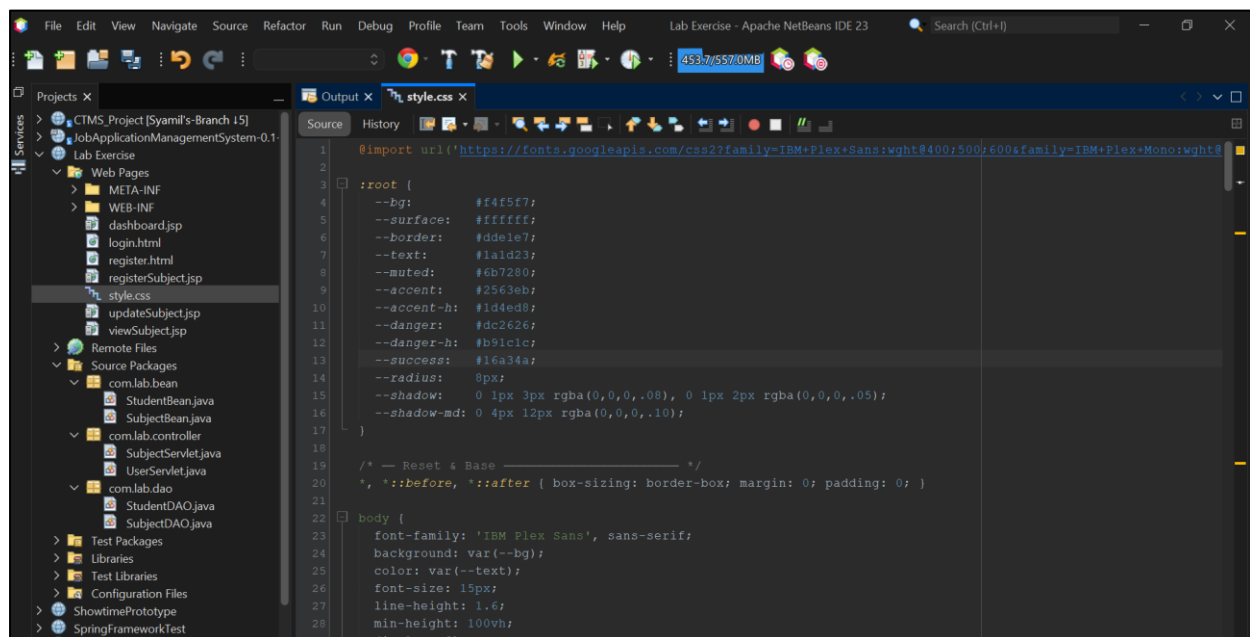


```

/* — Responsive — */
@media (max-width: 480px) {
  .card { padding: 28px 20px; }
  th, td { padding: 10px 14px; }
  .table-header { padding: 16px 14px 10px; }
}

```

Netbean



Source Code (Java):

StudentBean.java

```
package com.lab.bean;

public class StudentBean implements java.io.Serializable {

    private String matricNo;
    private String fullname;
    private String password;
    private String base64Image;

    public StudentBean() {
    }

    public String getMatricNo() {
        return matricNo;
    }

    public void setMatricNo(String matricNo) {
        this.matricNo = matricNo;
    }

    public String getFullname() {
        return fullname;
    }

    public void setFullname(String fullname) {
        this.fullname = fullname;
    }

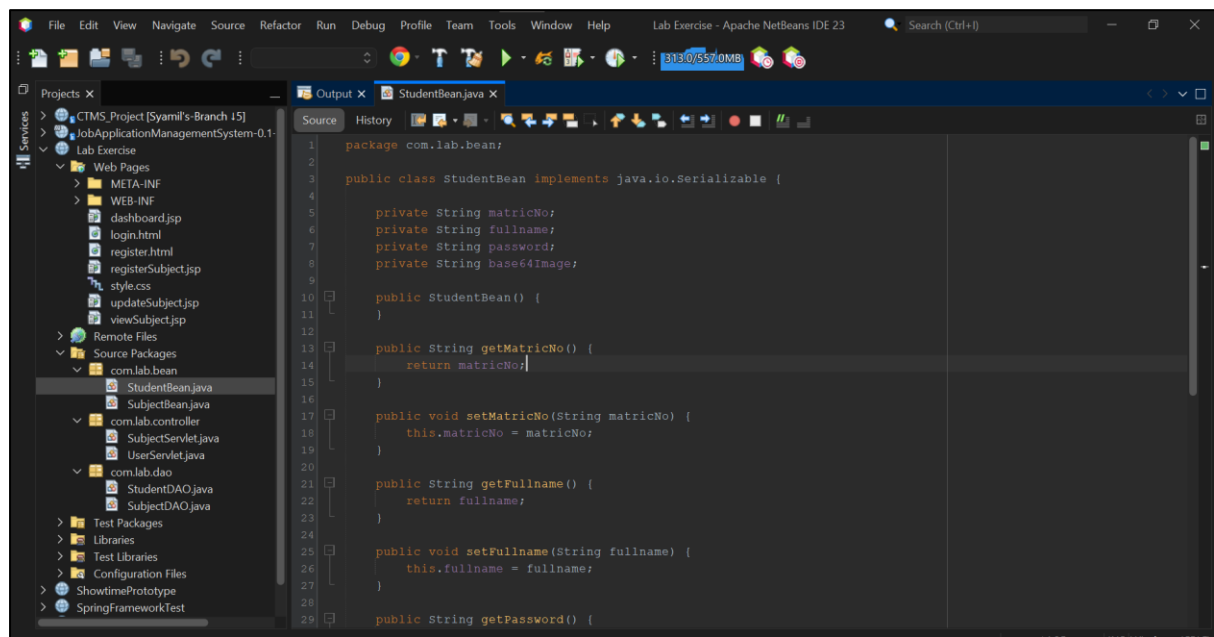
    public String getPassword() {
        return password;
    }

    public void setPassword(String password) {
        this.password = password;
    }

    public String getBase64Image() {
        return base64Image;
    }

    public void setBase64Image(String base64Image) {
        this.base64Image = base64Image;
    }
}
```

Netbean



SubjectBean.java

```

package com.lab.bean;

public class SubjectBean {

    private int id;
    private String subjectName;
    private String matricNo;

    public SubjectBean() {
    }
    public SubjectBean(int id, String subjectName, String matricNo)
    {
        this.id = id;
        this.subjectName = subjectName;
        this.matricNo = matricNo;
    }

    public String getMatricNo() {
        return matricNo;
    }

    public void setMatricNo(String matricNo) {
        this.matricNo = matricNo;
    }

    public String getSubjectName() {

```

```

        return subjectName;
    }

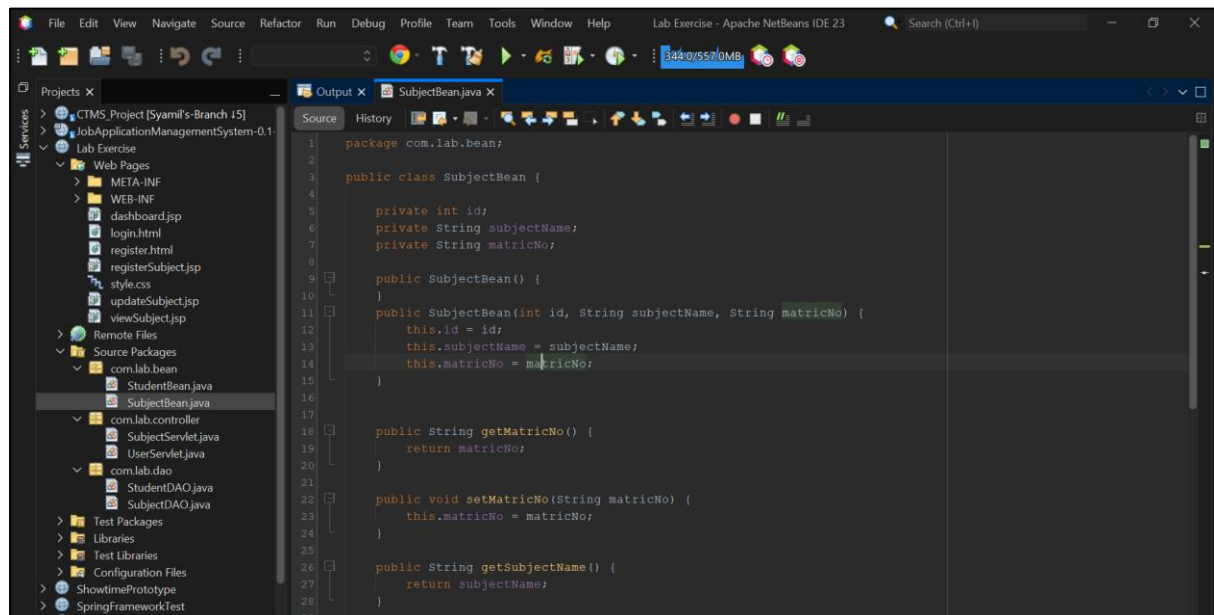
    public void setSubjectName(String subjectName) {
        this.subjectName = subjectName;
    }

    public int getId() {
        return id;
    }

    public void setId(int id) {
        this.id = id;
    }
}

```

Netbean



SubjectServlet.java

```

package com.lab.controller;

import com.lab.bean.StudentBean;
import com.lab.bean.SubjectBean;
import com.lab.dao.SubjectDAO;
import jakarta.servlet.RequestDispatcher;
import jakarta.servlet.ServletException;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

```

```

import java.io.IOException;
import java.util.ArrayList;
import java.util.logging.Level;
import java.util.logging.Logger;

public class SubjectServlet extends HttpServlet {

    private SubjectDAO subjectDAO = new SubjectDAO();

    @Override
    protected void doPost(HttpServletRequest request,
        HttpServletResponse response) throws IOException {
        String action = request.getParameter("action");

        if (action.equals("enroll")) {
            StudentBean student = (StudentBean)
request.getSession().getAttribute("loggedUser");
            String subjectName =
request.getParameter("subjectName");
            SubjectBean subject = new SubjectBean();
            subject.setMatricNo(student.getMatricNo());
            subject.setSubjectName(subjectName);

            try {
                subjectDAO.registerSubject(subject);
            } catch (Exception ex) {

Logger.getLogger(SubjectServlet.class.getName()).log(Level.SEVERE,
null, ex);
            }

            response.sendRedirect("viewSubject.jsp");
        } else if (action.equals("delete")) {
            StudentBean student = (StudentBean)
request.getSession().getAttribute("loggedUser");
            int subjectId =
Integer.parseInt(request.getParameter("subjectId"));
            System.out.println("ID: "+subjectId);
            try {

System.out.println(subjectDAO.deleteSubject(subjectId));
            } catch (Exception ex) {

Logger.getLogger(SubjectServlet.class.getName()).log(Level.SEVERE,
null, ex);
            }

            response.sendRedirect("viewSubject.jsp");
        } else if (action.equals("update")) {

```

```

        StudentBean student = (StudentBean)
request.getSession().getAttribute("loggedUser");
        String subjectName =
request.getParameter("subjectName");
        int subjectId =
Integer.parseInt(request.getParameter("id"));
        SubjectBean subject = new SubjectBean();
        subject.setMatricNo(student.getMatricNo());
        subject.setSubjectName(subjectName);
        subject.setId(subjectId);

        try {
            subjectDAO.updateSubject(subject);
        } catch (Exception ex) {

Logger.getLogger(SubjectServlet.class.getName()).log(Level.SEVERE,
null, ex);

        }

        response.sendRedirect("viewSubject.jsp");
    }

}

@Override
protected void doGet(HttpServletRequest request,
HttpServletResponse response) throws IOException, ServletException {
    StudentBean student = (StudentBean)
request.getSession().getAttribute("loggedUser");
    String action = request.getParameter("action");

    if (action.equals("view")) {
        try {
            ArrayList<SubjectBean> subjects =
subjectDAO.getAllSubjectByMatricNo(student.getMatricNo());

            request.setAttribute("subjects", subjects);
            RequestDispatcher requestDispatcher =
request.getRequestDispatcher("viewSubject.jsp");
            requestDispatcher.forward(request, response);
        } catch (Exception ex) {

Logger.getLogger(SubjectServlet.class.getName()).log(Level.SEVERE,
null, ex);

        }

        return;
    } else if (action.equals("viewUpdate")) {

```

```

        int id =
Integer.parseInt(request.getParameter("subjectId"));
        SubjectBean subject = null;
        try {
            subject = subjectDAO.getSubjectById(id);
        } catch (Exception ex) {

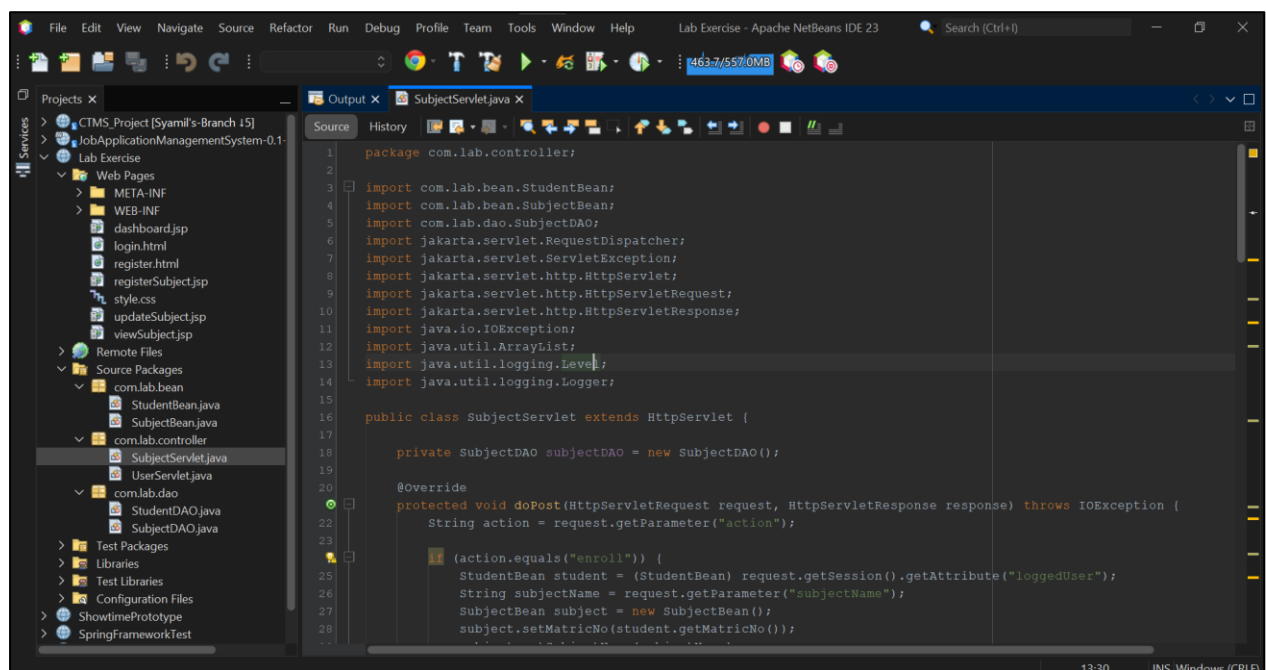
Logger.getLogger(SubjectServlet.class.getName()).log(Level.SEVERE,
null, ex);

        }
        request.setAttribute("subject", subject);
        System.out.println(subject);
        RequestDispatcher requestDispatcher =
request.getRequestDispatcher("updateSubject.jsp");
        requestDispatcher.forward(request, response);
        return;
    }

}
}

```

Netbean



UserServlet.java

```
package com.lab.controller;

import com.lab.bean.StudentBean;
import com.lab.dao.StudentDAO;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.MultipartConfig;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
import jakarta.servlet.http.Part;
import java.io.IOException;
import java.io.InputStream;

@MultipartConfig(maxFileSize = 16177215) // max 16 MB required for
image upload
public class UserServlet extends HttpServlet {

    private StudentDAO studentDAO = new StudentDAO();

    @Override
    protected void doPost(HttpServletRequest request,
HttpServletResponse response) throws IOException, ServletException {
        response.setContentType("text/html;charset=UTF-8");

        String action = request.getParameter("action");
        if ("register".equals(action)) {

            StudentBean newStudent = new StudentBean();

newStudent.setMatricNo(request.getParameter("matricNo"));

newStudent.setFullname(request.getParameter("fullname"));

newStudent.setPassword(request.getParameter("password"));
            InputStream inputStream = null;
            Part filePart = request.getPart("profileImage");
            if (filePart != null && filePart.getSize() > 0) {
                inputStream = filePart.getInputStream();
            }
            boolean isSuccess =
studentDAO.registerStudent(newStudent, inputStream);
            if (isSuccess) {
                response.getWriter().println("Registration
Successful!<a href ='login.html'>Login here< / a >");
            } else {
                response.getWriter().println("Registration
```



```

Failed!");
    }
    } else if ("login".equals(action)) {

        String matric = request.getParameter("matricNo");
        String pass = request.getParameter("password");
        StudentBean student = studentDAO.loginStudent(matric,
pass);

        if (student != null) {

            HttpSession session = request.getSession();
            session.setAttribute("loggedUser", student);
            response.sendRedirect("viewSubject.jsp");
        } else {
            response.getWriter().println("Invalid Credentials!<a
href=\"login.html\">Try Again</a>");
        }
    }
}

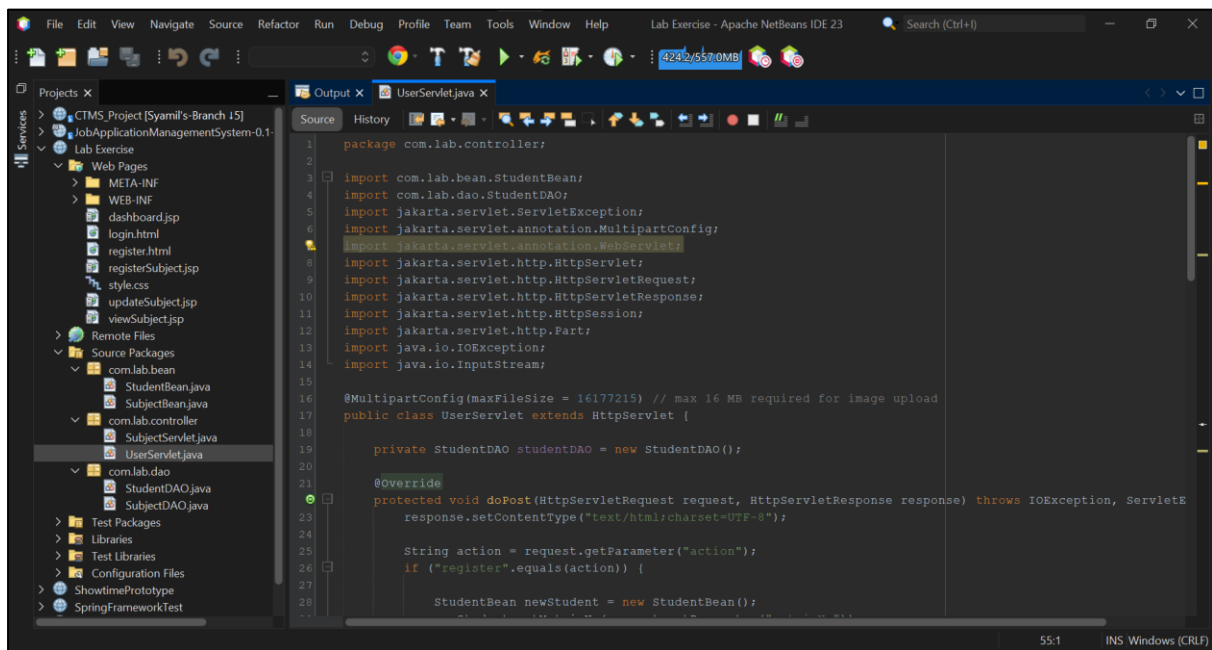
@Override
protected void doGet(HttpServletRequest request,
HttpServletResponse response)
    throws ServletException, IOException {
    String action = request.getParameter("action");
    HttpSession session = request.getSession(false);
    if ("logout".equals(action)) {

        if (session != null) {
            session.invalidate();
        }
        response.sendRedirect("login.html");
    } else if ("delete".equals(action)) {

        if (session != null &&
session.getAttribute("loggedUser") != null) {
            StudentBean user = (StudentBean)
session.getAttribute("loggedUser");
            studentDAO.deleteStudent(user.getMatricNo());
            session.invalidate(); // Destroy session after
account deletion
        }
        response.sendRedirect("register.html");
    }
}
}

```

Netbean



StudentDAO.java

```
package com.lab.dao;

import com.lab.bean.StudentBean;
import java.io.ByteArrayOutputStream;
import java.io.InputStream;
import java.sql.Blob;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.util.Base64;

public class StudentDAO {

    private Connection getConnection() throws Exception {
        Class.forName("com.mysql.cj.jdbc.Driver");
        return
DriverManager.getConnection("jdbc:mysql://localhost:3307/lab7_db",
"root", "");
    }

    public boolean registerStudent(StudentBean student, InputStream
imageStream) {
        try (Connection conn = getConnection()) {
            String sql =
```

```

        "INSERT INTO students (matric_no, password, fullname,
profile_image) VALUES(?, ?, ?, ?)";
        PreparedStatement pstmt = conn.prepareStatement(sql);
        pstmt.setString(1, student.getMatricNo());
        pstmt.setString(2, student.getPassword());
        pstmt.setString(3, student.getFullname());
        pstmt.setBlob(4, imageStream); // Set BLOB data
        return pstmt.executeUpdate() > 0;
    } catch (Exception e) {
        e.printStackTrace();
        return false;
    }
}

public StudentBean loginStudent(String matricNo, String
password) {
    StudentBean student = null;
    try (Connection conn = getConnection()) {
        String sql = "SELECT * FROM students WHERE matric_no = ?
AND password = ?";
        PreparedStatement pstmt = conn.prepareStatement(sql);
        pstmt.setString(1, matricNo);

        pstmt.setString(2, password);
        ResultSet rs = pstmt.executeQuery();
        if (rs.next()) {
            student = new StudentBean();
            student.setMatricNo(rs.getString("matric_no"));
            student.setFullname(rs.getString("fullname"));

            Blob blob = rs.getBlob("profile_image");
            if (blob != null) {
                InputStream inputStream =
blob.getBinaryStream();
                ByteArrayOutputStream outputStream = new
ByteArrayOutputStream();
                byte[] buffer = new byte[4096];
                int bytesRead = -1;
                while ((bytesRead = inputStream.read(buffer)) !=
-1) {
                    outputStream.write(buffer, 0, bytesRead);
                }
                byte[] imageBytes = outputStream.toByteArray();
                student.setBase64Image(Base64.getEncoder().encodeToString(imageBytes
));
            }
        }
    } catch (Exception e) {
        e.printStackTrace();
    }
}

```

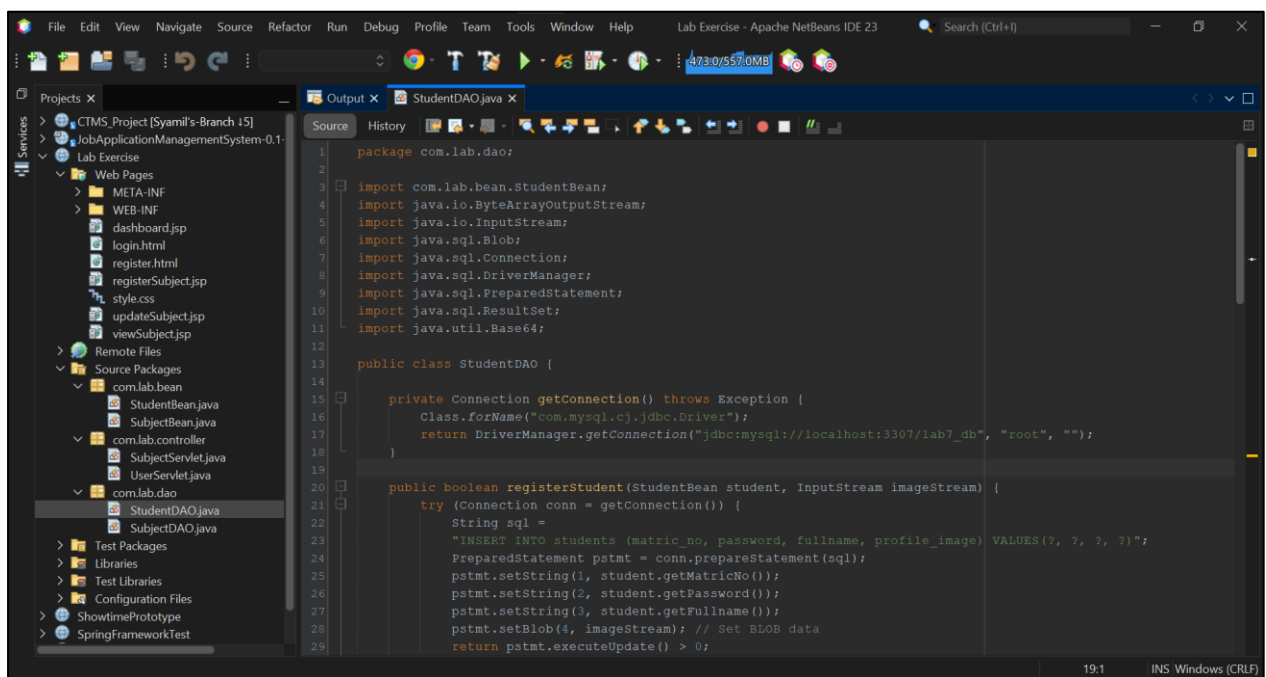
```

    }
    return student;
}

public boolean deleteStudent(String matricNo) {
    try (Connection conn = getConnection()) {
        String sql = "DELETE FROM students WHERE matric_no = ?";
        PreparedStatement pstmt = conn.prepareStatement(sql);
        pstmt.setString(1, matricNo);
        return pstmt.executeUpdate() > 0;
    } catch (Exception e) {
        e.printStackTrace();
        return false;
    }
}
}

```

Netbean



SubjectDAO.java

```
package com.lab.dao;

import com.lab.bean.SubjectBean;
import java.sql.*;
import java.util.ArrayList;

public class SubjectDAO {

    private Connection getConnection() throws Exception {
        Class.forName("com.mysql.cj.jdbc.Driver");
        return
DriverManager.getConnection("jdbc:mysql://localhost:3307/lab7_db",
"root", "");
    }

    public boolean registerSubject(SubjectBean subject) throws
Exception {
        String sql = "INSERT INTO registered_subjects(subject_name,
matric_no) VALUE(?, ?)";

        String matricNo = subject.getMatricNo();
        if (matricNo == null) {
            return false;
        }

        try (Connection conn = getConnection()) {
            try (PreparedStatement pstmt =
conn.prepareStatement(sql)) {
                pstmt.setString(1, subject.getSubjectName());
                pstmt.setString(2, matricNo);

                return 0 > pstmt.executeUpdate();
            }
        }
    }

    public boolean updateSubject(SubjectBean subject) throws
Exception {
        String sql = "UPDATE registered_subjects SET subject_name =
? WHERE id = ?";

        try (Connection conn = getConnection()) {
            try (PreparedStatement pstmt =
conn.prepareStatement(sql)) {
                pstmt.setString(1, subject.getSubjectName());
                pstmt.setInt(2, subject.getId());

                int rows = pstmt.executeUpdate();
            }
        }
    }
}
```

```

        return rows > 0;
    }
}

public boolean deleteSubject(int id) throws Exception {
    String sql = "DELETE FROM registered_subjects WHERE id = ?";

    try (Connection conn = getConnection()) {
        try (PreparedStatement pstmt =
conn.prepareStatement(sql)) {
            pstmt.setInt(1, id);
            System.out.println(id);
            return 0 > pstmt.executeUpdate();
        }
    }

    public ArrayList<SubjectBean> getAllSubjectByMatricNo(String
matricNo) throws SQLException, Exception {
        String sql = "SELECT * FROM registered_subjects WHERE
matric_no = ?";

        try (Connection conn = getConnection()) {
            try (PreparedStatement pstmt =
conn.prepareStatement(sql)) {
                pstmt.setString(1, matricNo);

                try (ResultSet result = pstmt.executeQuery()) {
                    ArrayList<SubjectBean> subjects = new
ArrayList<SubjectBean>();
                    while (result.next()) {
                        subjects.add(
                            new SubjectBean(
                                result.getInt("id"),
                                result.getString("subject_name"),
                                result.getString("matric_no")
                            )
                        );
                    }
                    return subjects;
                }
            }
        }

        public SubjectBean getSubjectById(int id) throws SQLException,

```

```

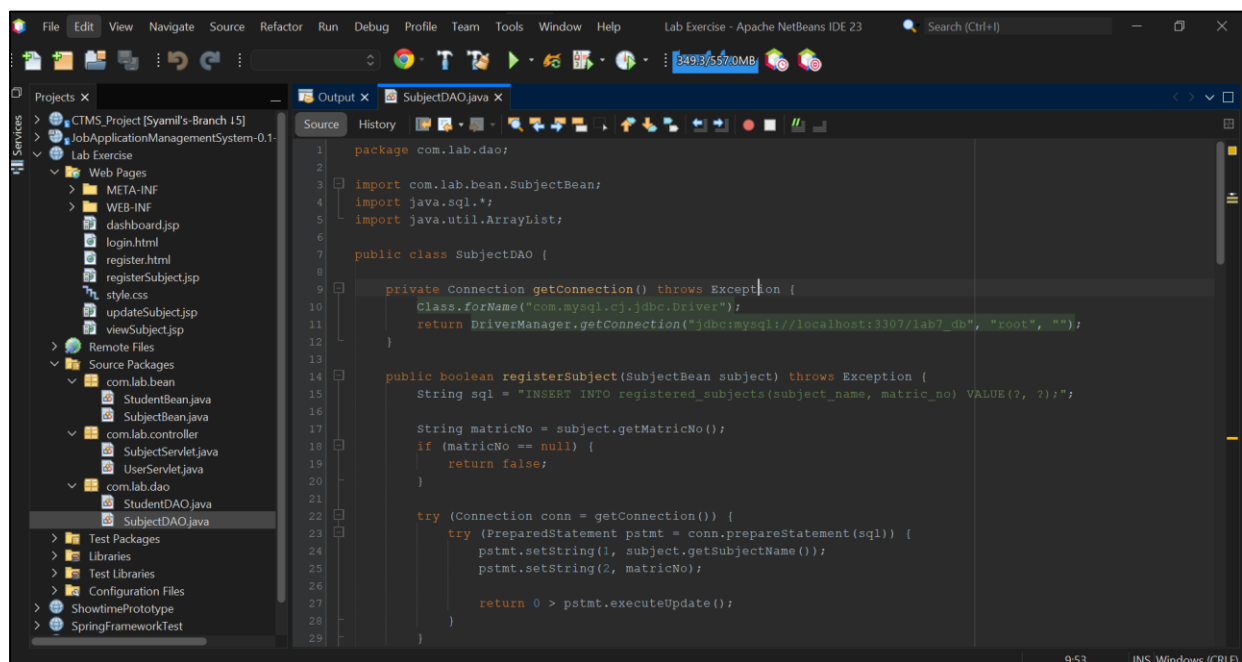
Exception {
    String sql = "SELECT * FROM registered_subjects WHERE id =
?";

    try (Connection conn = getConnection()) {
        try (PreparedStatement pstmt =
conn.prepareStatement(sql)) {
            pstmt.setInt(1, id);

            try (ResultSet result = pstmt.executeQuery()) {
                if (result.next()) {
                    return new SubjectBean(
                        result.getInt("id"),
                        result.getString("subject_name"),
                        result.getString("matric_no")
                    );
                }
            }
        }
    }
    return null;
}
}

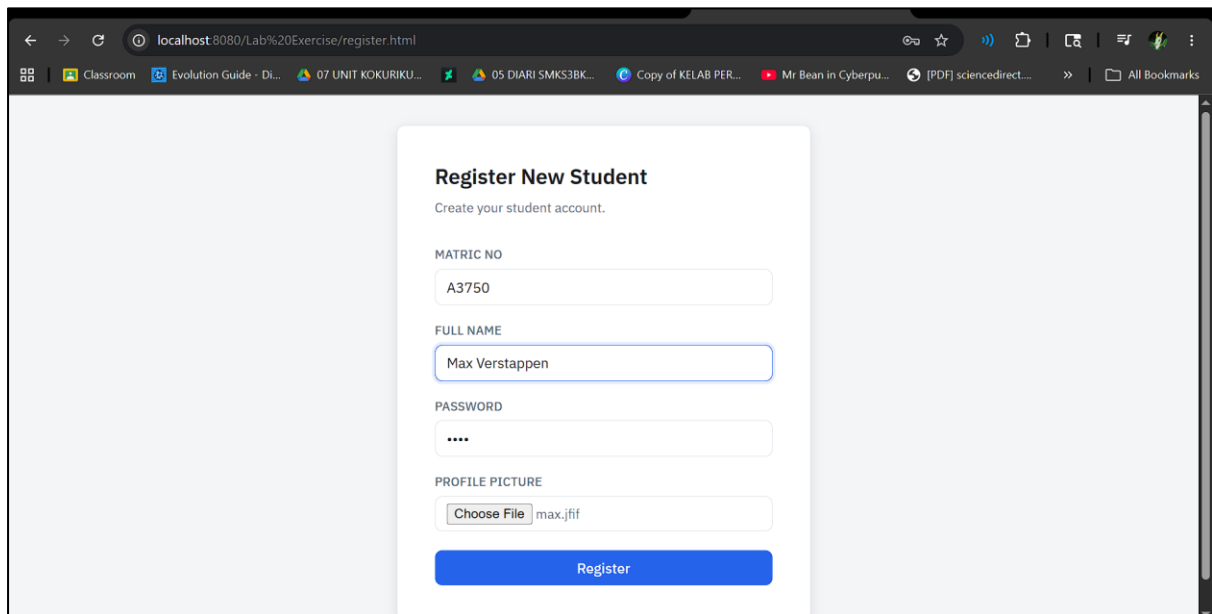
```

Netbean



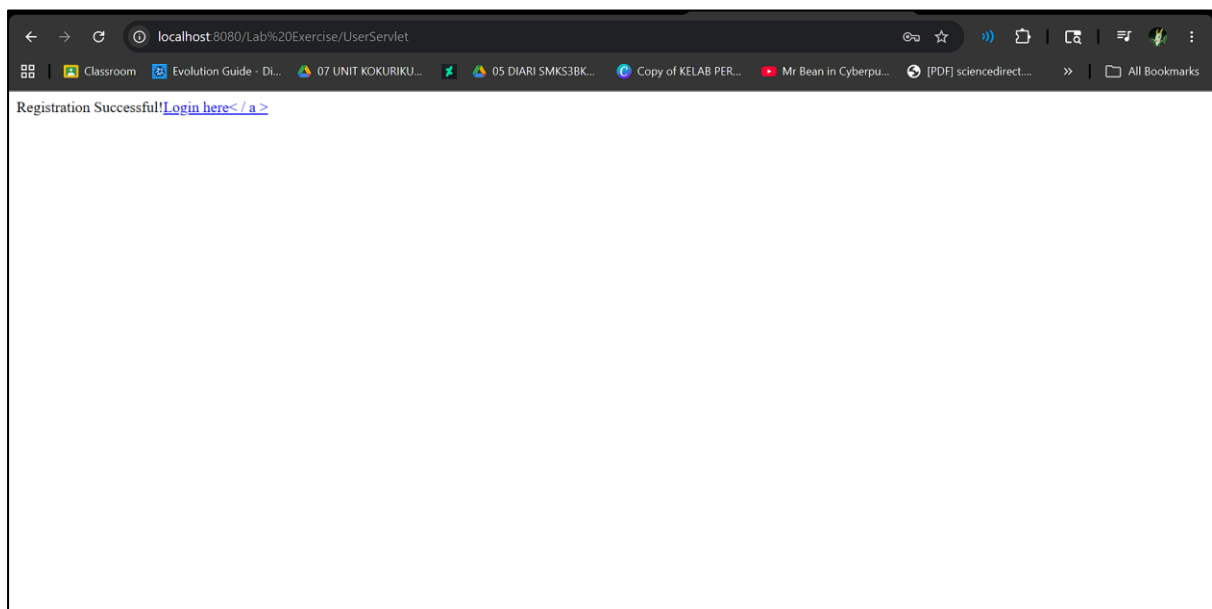
Output:

Register Student Page

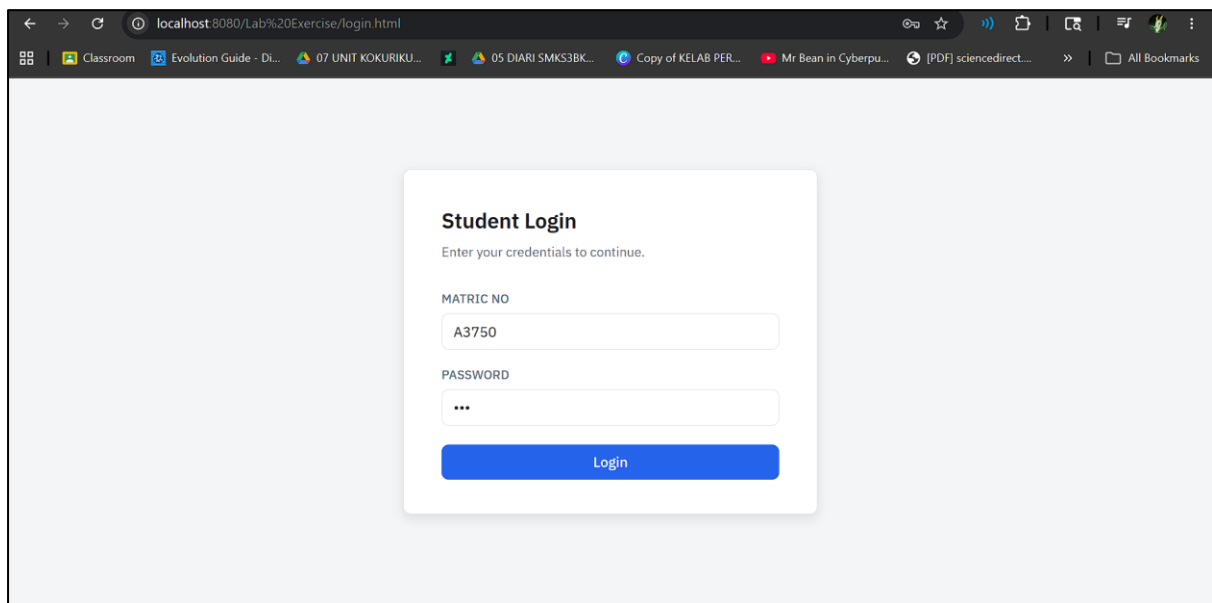


A screenshot of a web browser displaying the 'Register New Student' page. The browser's address bar shows 'localhost:8080/Lab%20Exercise/register.html'. The page features a central white card with the title 'Register New Student' and the subtitle 'Create your student account.' Below this, there are four input fields: 'MATRIC NO' with the value 'A3750', 'FULL NAME' with the value 'Max Verstappen', 'PASSWORD' with masked characters '....', and 'PROFILE PICTURE' with a 'Choose File' button and the filename 'max.jif'. A blue 'Register' button is positioned at the bottom of the card. The browser's tab bar shows several open tabs, including 'Classroom', 'Evolution Guide - Di...', '07 UNIT KOKURIKU...', '05 DIARI SMKS3BK...', 'Copy of KELAB PER...', 'Mr Bean in Cyberpu...', and '[PDF] sciencedirect...'. The bookmarks bar shows 'All Bookmarks'.

Registration Feedback Page

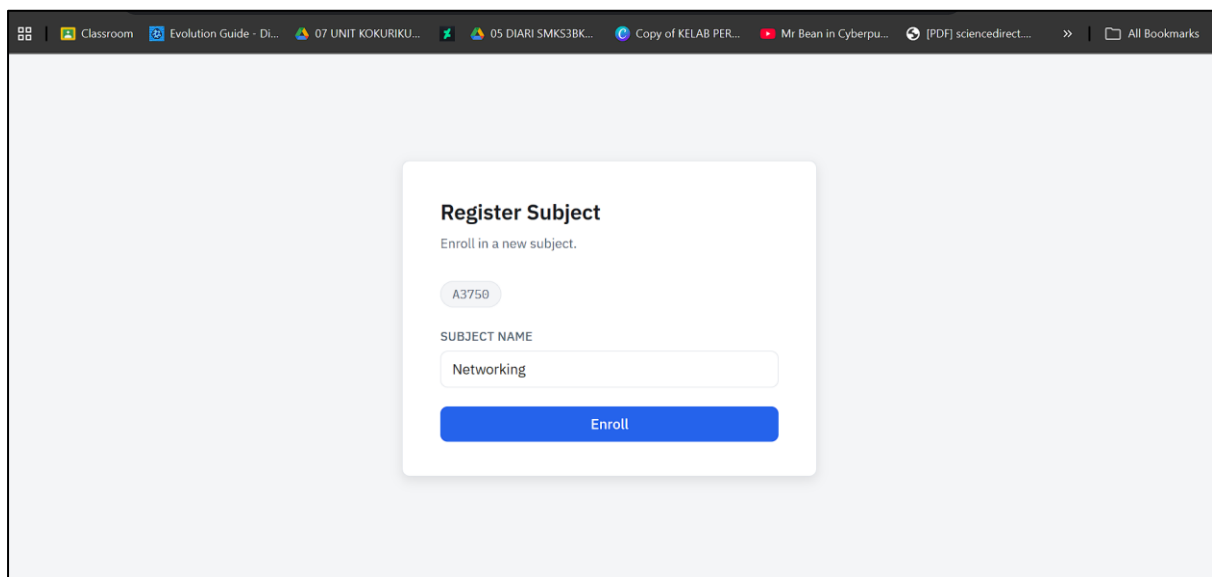


Student Login Page



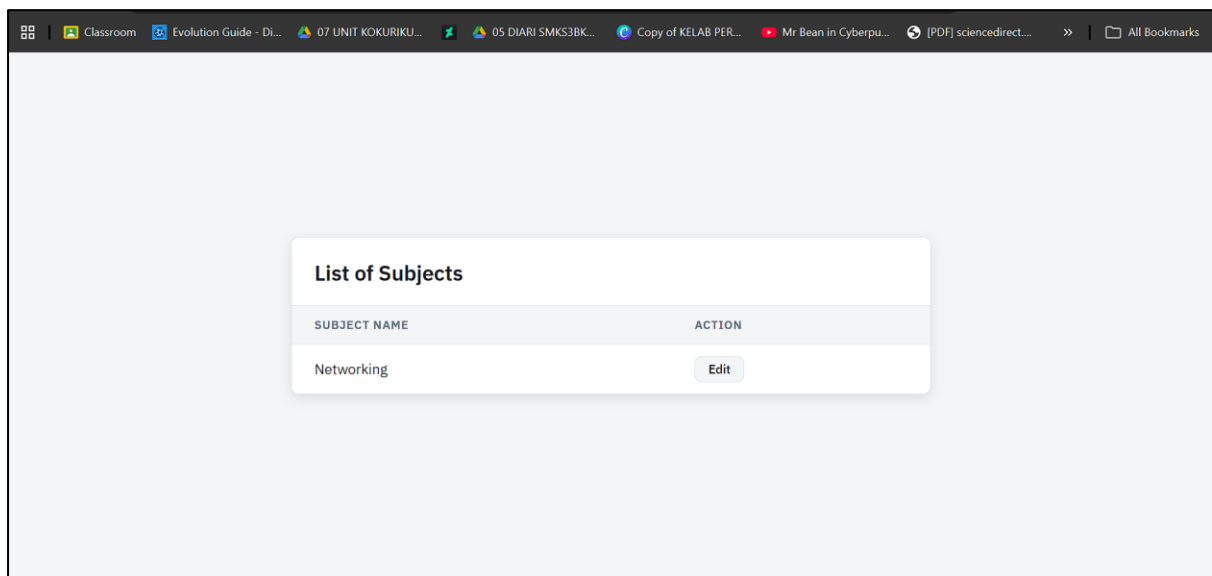
The screenshot shows a web browser window with the address bar displaying `localhost:8080/Lab%20Exercise/login.html`. The browser's tab bar includes several tabs: Classroom, Evolution Guide - Di..., 07 UNIT KOKURIKU..., 05 DIARI SMKS3BK..., Copy of KELAB PER..., Mr Bean in Cyberpu..., [PDF] sciencedirect..., and All Bookmarks. The main content area features a white login form titled "Student Login" with the instruction "Enter your credentials to continue." The form contains two input fields: "MATRIC NO" with the value "A3750" and "PASSWORD" with masked characters "...". A blue "Login" button is positioned at the bottom of the form.

Register Subject Page

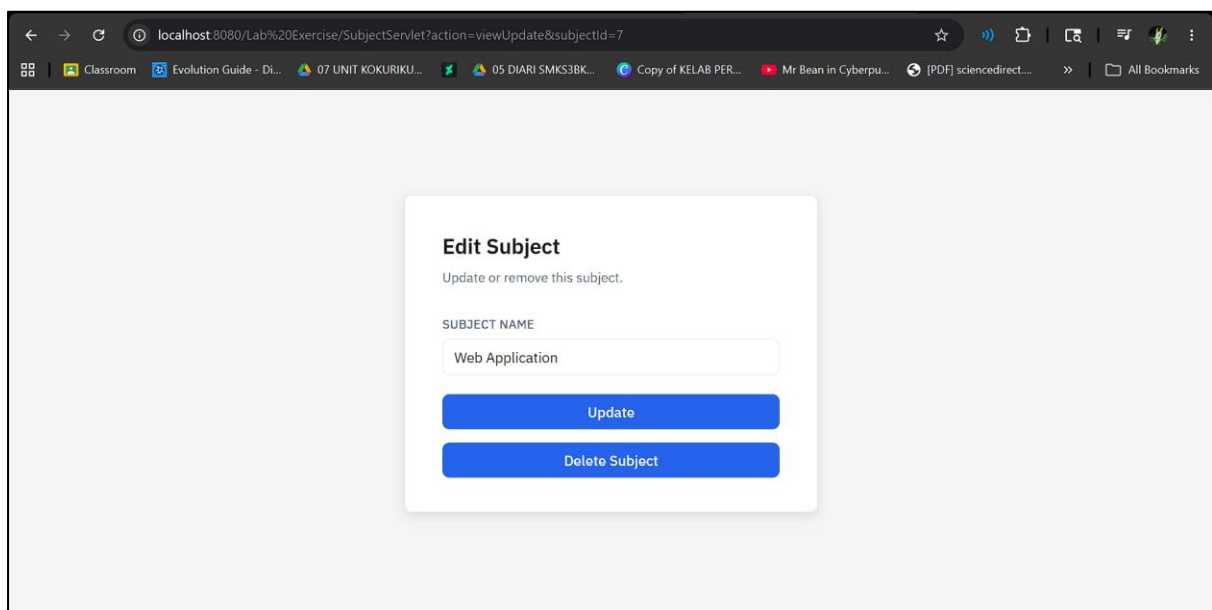


The screenshot shows a web browser window with the same address bar and tab bar as the previous page. The main content area features a white registration form titled "Register Subject" with the instruction "Enroll in a new subject." The form includes a "MATRIC NO" field with the value "A3750" displayed in a light gray box. Below this is a "SUBJECT NAME" input field containing the text "Networking". A blue "Enroll" button is located at the bottom of the form.

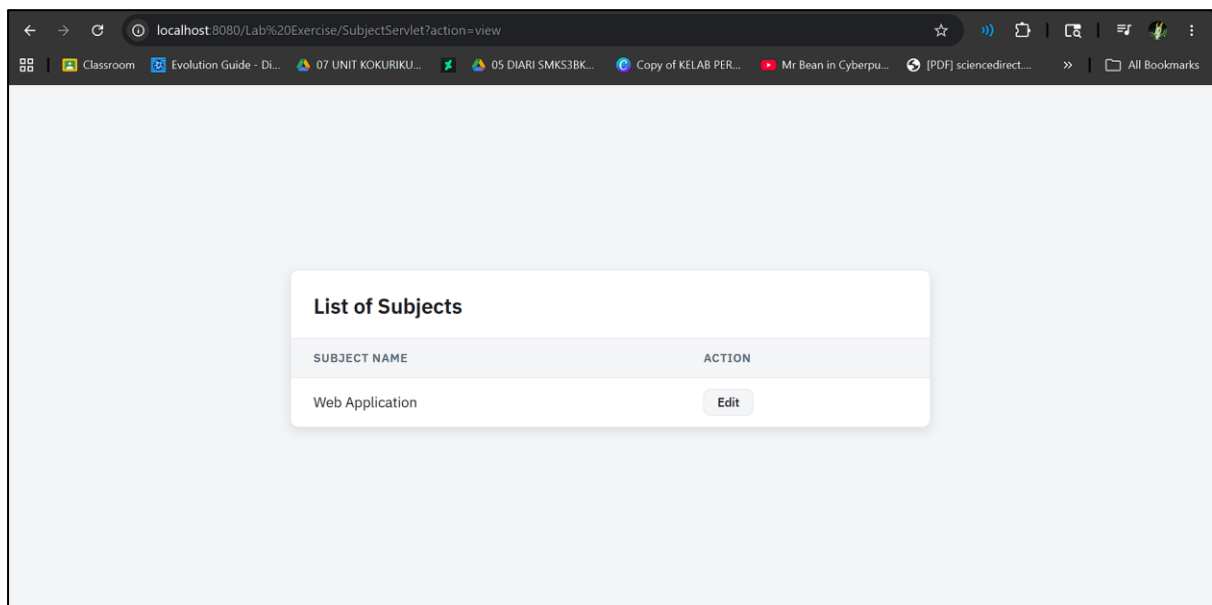
List of Subjects Page



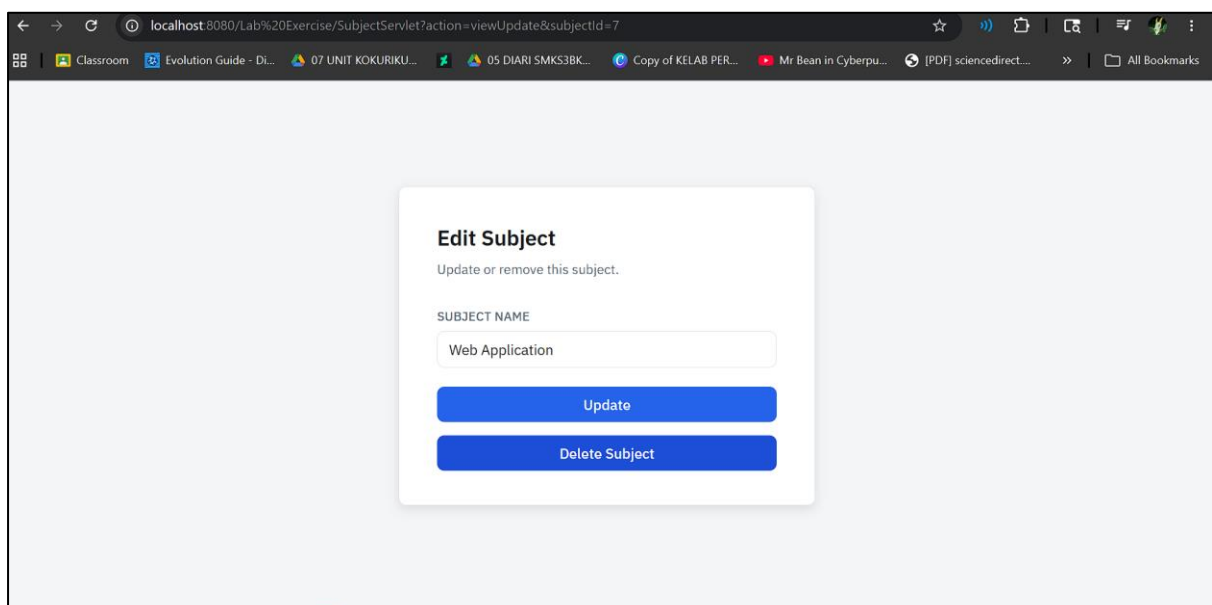
Edit Subject Page (Networking to Web Application)



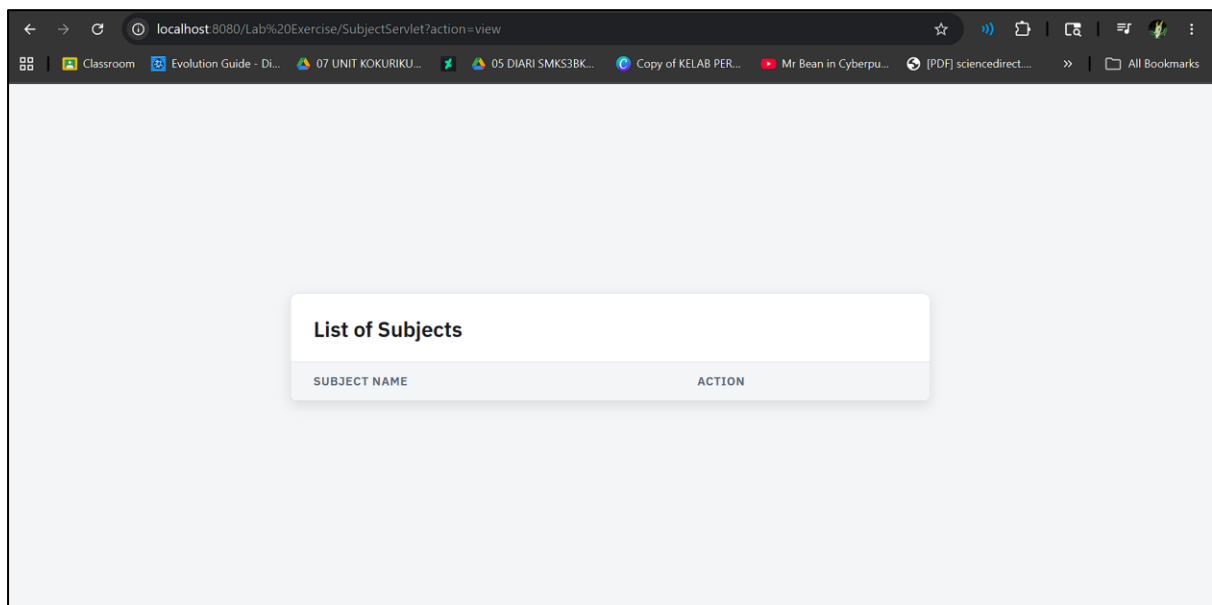
After Update



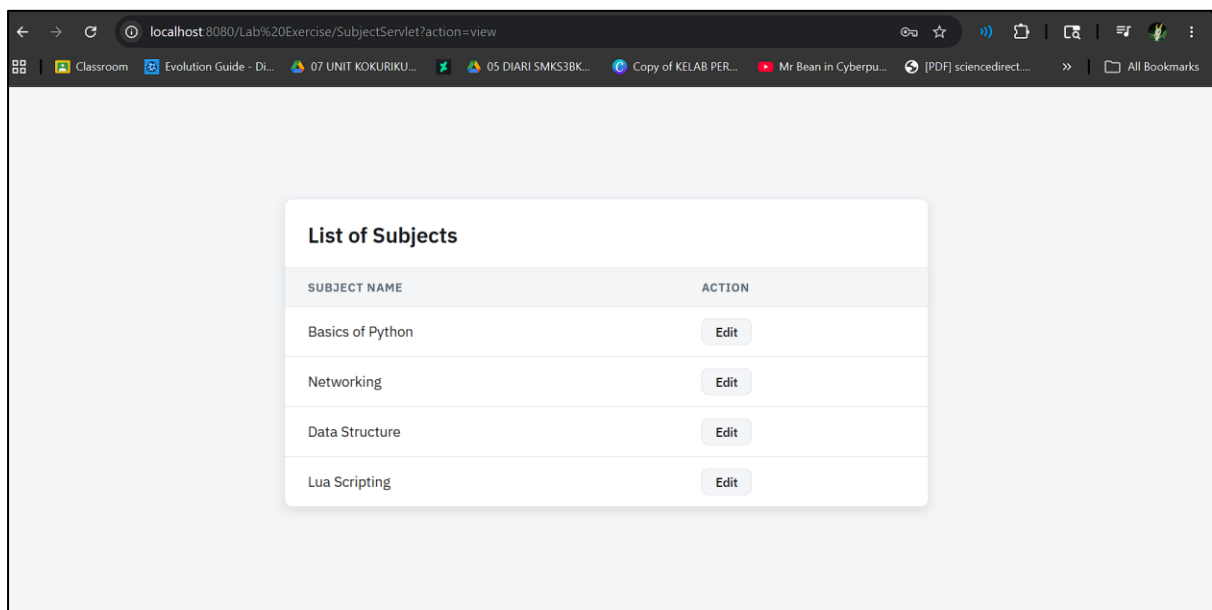
Delete Page



After Delete



Login with Another Account



Updated MySQL PHP MyAdmin

Server: 127.0.0.1:3307 > Database: lab7_db > Table: students

Showing rows 0 - 1 (2 total, Query took 0.0006 seconds.)

```
SELECT * FROM `students`
```

Number of rows: 25 Filter rows: Search this table Sort by key: None

	id	matric_no	password	fullname	profile_image
☐	1	S76064	123	Syamil	[BLOB - 8.4 KiB]
☐	2	A3750	1234	Max Verstappen	[BLOB - 5.5 KiB]

Query results operations: Print, Copy to clipboard, Export, Display chart, Create view

Server: 127.0.0.1:3307 > Database: lab7_db > Table: registered_subjects

Showing rows 0 - 4 (5 total, Query took 0.0009 seconds.)

```
SELECT * FROM `registered_subjects`
```

Number of rows: 25 Filter rows: Search this table Sort by key: None

	id	subject_name	matric_no
☐	2	Basics of Python	S76064
☐	4	Networking	S76064
☐	5	Data Structure	S76064
☐	6	Lua Scripting	S76064
☐	8	Automotive Engineering with C++	A3750